MANYARA REGION
FORM FOUR MOCK EXAMINATION 2021
032/2B **CHEMISTRY 2B**
ACTUAL PRACTICAL B
MARKING SCHEME

1. Table1: Experimental results

Experiments	Pilot	1	2	2
Final reading	13.20	20.20	37.20	49.20
Initial reading	1.00	13.20	25.20	37.20
Volume acid used	12.20	12.00	12.00	12.00

Pipette used was $25cm^3$ **(04 Marks)**

Note: if $20.00cm^3$ pipette was used the volume of acid will be $9.60cm^3$

(a) (i) The volume H_2SO_4 required by $25cm^3$ of sodium hydroxide

The volume of H_2SO_4 = $\frac{V_1 + V_2 + V_3}{3}$ **(01 mark)**

= $\frac{12.00cm^3 + 12.00cm^3 + 12.00cm^3}{3}$ **(01 mark)**

= $\frac{36}{3}$

= $12.00cm^3$

Therefore the mean titre volume is = $12.00cm^3$ **(01 mark)**

(ii) Find concentration of sodium hydroxide NaOH

From

0.96g = $250cm^3$

x = $1000cm^3$ **(01 mark)**

$X = \frac{0.96 \times 1000}{250}$ **(01 mark)**

= $3.84g/dm^3$

Therefore the concentration of NaOH is $3.84g/dm^3$ **(02 marks)**

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(iii) **Find molarity of sodium hydroxide**

Data given:

Concentration of NaOH = $3.84g/dm^3$

Molar mass of NaOH = 40g/mol

Molarity of NaOH =?

From

$$\text{Molarity} = \frac{\text{Concentration}}{\text{Molar mass}} \quad \textbf{(01 mark)}$$

$$= \frac{3.84g/dm^3}{40g/mol} \quad \textbf{(01 mark)}$$

$$= 0.096M$$

Therefore the molarity of NaOH is 0.096M (02 marks)

(b) (i) **Find the number of moles of sulphuric acid.**

Data given:

Molarity of H_2SO_4 = 0.1M

Volume of H_2SO_4 = $12cm^3$ = $0.012dm^3$

Number of moles of H_2SO_4 = ?

From

$$\frac{\text{Molarity}}{1} = \frac{\text{number of moles}}{\text{Volume of a solution } (dm^3)} \quad \textbf{(01 mark)}$$

Number of moles = Molarity x volume **(01 mark)**

= 0.1 x 0.012

0.0012moles **(01 mark)**

Therefore the number of moles of H_2SO_4 acid is 0.0012 moles **(02 marks)**

(ii) **Find the number of moles NaOH.**

Data given:

Molarity of **NaOH** = 0.096M

Volume of **NaOH** = $25cm^3$ = $0.025dm^3$

Number of moles of **NaOH** = ?

From

$$\frac{\text{Molarity}}{1} = \frac{\text{number of moles}}{\text{Volume of a solution } (dm^3)} \quad \textbf{(01 mark)}$$

Number of moles = Molarity x volume **(01 mark)**

= 0.096 x 0.025 **(01 mark)**

= 0.0024moles

Therefore the number of moles of **NaOH** is 0.0024 moles **(02 marks)**

2. (a) Table 2: Experimental results

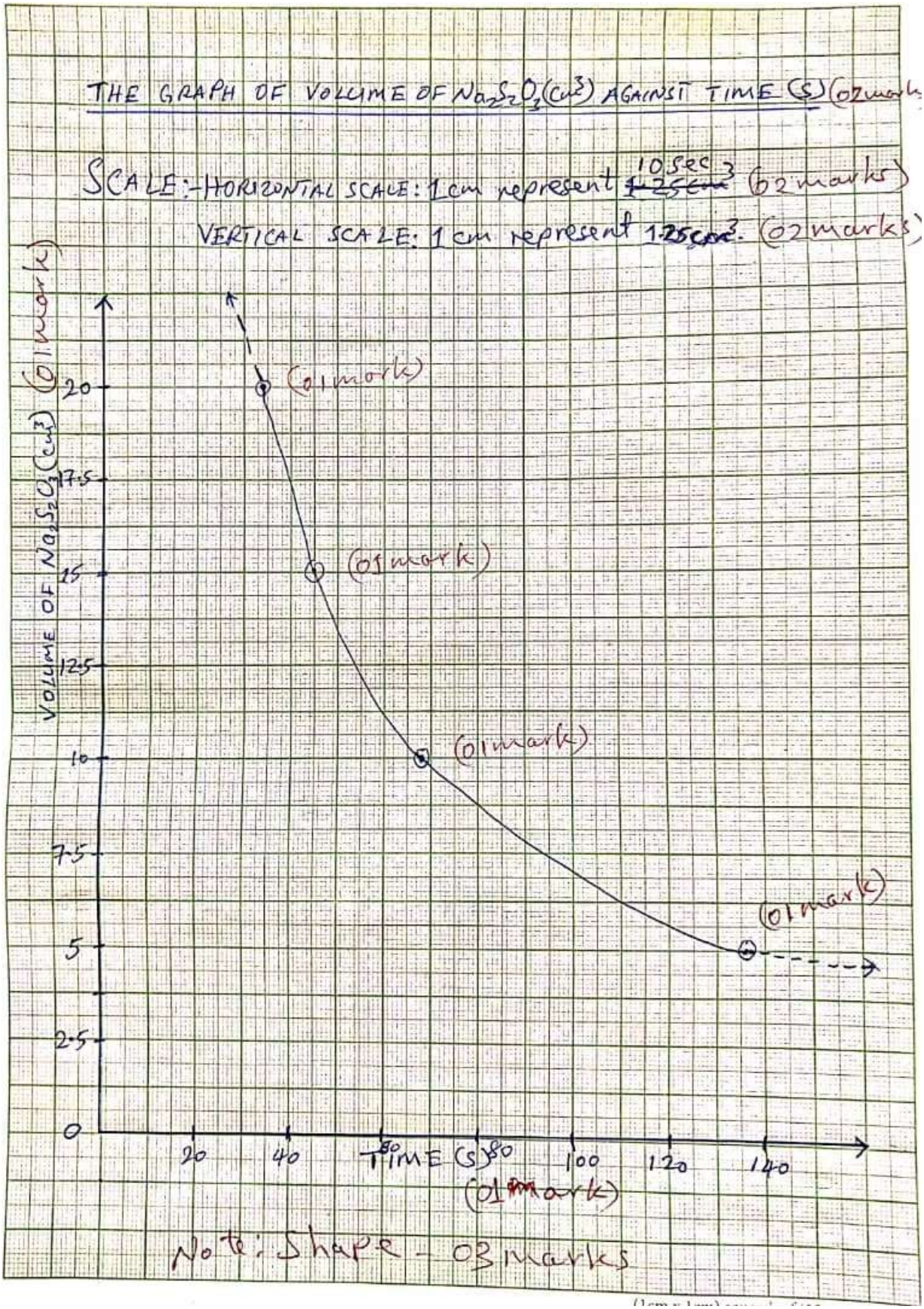
Volume of solution **Q** (cm^3)	Volume of Distilled water (cm^3)	Volume of solution **Z** (cm^3)	Time (i) for the cross (x) to disappear	Rate of reaction 1/t (s^{-1})
20	0	10	34	0.029
15	5	10	45	0.022
10	10	10	68	0.015
5	15	10	136	0.007

(04 marks)

(b) The equation between **Z** and **Q**

$Na_2S_2O_{3\,(aq)} + 2HCl_{(aq)} \longrightarrow 2NaCl_{(aq)} + H_2O_{(l)} + SO_{2(g)} + S_{(s)}$ **(03 marks)**

(c) The graph of sodium thiosulphate (cm^3) against time (sec)


THE GRAPH OF VOLUME OF $Na_2S_2O_3$ (cm^3) AGAINST TIME (S) (02 marks)
SCALE:- HORIZONTAL SCALE: 1cm represent 10 sec (02 marks)
VERTICAL SCALE: 1 cm represent 1.25 cm^3. (02 marks)
VOLUME OF $Na_2S_2O_3$ (cm^3) (01 mark)
20
17.5
15
12.5
10
7.5
5
2.5
0
(01 mark)
(01 mark)
(01 mark)
(01 mark)
20
40
60
80
100
120
140
TIME (S)
(01 mark)
Note: Shape - 03 marks

(d) The graph shows that the rate of reaction decreases with an increase in time taken. Thus the rate of reaction is directly proportional to the concentration of the reacting substance but inversely proportional to the time taken. **(03 marks)**